Project Overview:

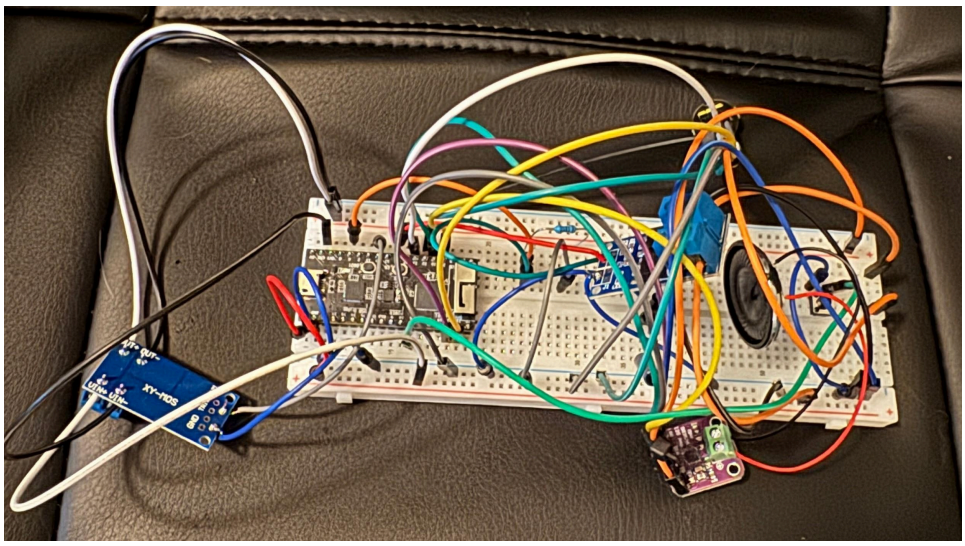
A voice message is recorded and it is played through the speaker when the PCB is moved.

The goal is to be able to have this as a flexible PCB which can be stuck to any item easily. To accomplish this it must be thin and small.

I have these different modules connected to an **ESP PICO V3** as shown below. Everything needs to be integrated onto a single PCB board. It should be made as small and thin as possible. This should not use any modules, it should use the SMD components.

The battery life must be at least 6 months.

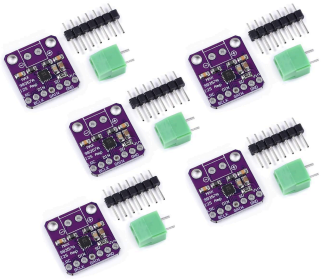
Prototype:



Modules and Wiring:

Max98357

LRC: GPIO25
BCLK: GPIO26
DIN: GPIO22
GAIN: GND
SD: NONE
GND: GND
Vin: Microphone MOSFET OUT+



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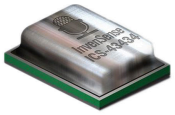
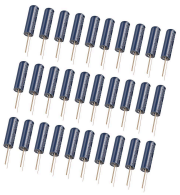
SD: GPIO33
VDD: Microphone MOSFET OUT+
GND: GND
SCK: GPIO32



WS: GPIO21
L/R: GND

SW-18010P Vibration Sensor



DO: GPIO4 (WITH ~4.7KΩ pull-up resistor)
GND: GND

1W 8 Ohm DIY Magnetic Speaker

Positive +: + Terminal on Max98357
Negiting -: - Terminal on Max98357

2 Pin Tactile Tact Push Button

COM: GND
NO: GPIO5

220mAh Battery

MOSFET for Speaker Power
Vin+: 3.3v
Vin-: GND
Out+: Speaker Vin
Out- GND
GND: GND
Trig: GPIO13

MOSFET for Microphone Power
Vin+: 3.3v
Vin-: GND
Out+: Microphone Vin
Out- GND
GND: GND
Trig: GPIO14